

LESSON 7-2 • SESSION 2 • PRACTICE • FORM A

Opposites of Rational Numbers

Learning target – I can find the opposite of a rational number and explain what taking the opposite twice does. • 6.NS.C.6a

REMEMBER

- **Opposites** are the same distance from 0 on different sides: 2.5 and -2.5 . Zero is its own opposite.
- $-(4.5)$ means **the opposite of 4.5**, so $-(4.5) = -4.5$ and $-(-4.5) = 4.5$.
- Taking the opposite **twice** gives back the number you started with.

WORKED EXAMPLEFind $-(-5.3)$. Then find the opposite of your answer.**Step 1** -5.3 is 5.3 units left of 0.**Step 2** Flip to the other side of 0, keeping the distance 5.3: $-(-5.3) = 5.3$.**Step 3** Flip again: the opposite of 5.3 is -5.3 .**Answer** 5.3, then -5.3 (two flips return to the start)**WORDS** **opposite** (*opuesto*) • **opposite of the opposite** (*opuesto del opuesto*) • **rational number** (*número racional*)**PRACTICE**

show your work • label units

1. Write the opposite of each number, or find the value.

a. -2.5 _____

b. $7/10$ _____

c. $-1/4$ _____

d. 0 _____

e. $-(4.5) =$ _____

f. $-(-4.5) =$ _____

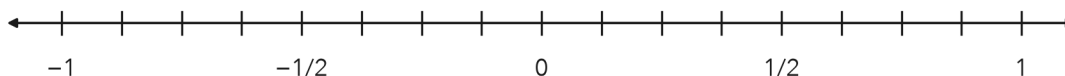
- 2.
- Scuba Dive Log.**
- A dive log lists four depths: reef
- -4.5
- m, ledge
- -7.25
- m, wreck
- -11.5
- m, sandbar
- -0.8
- m.

a. Write the opposite of each depth. _____

b. How far from 0 is the ledge depth? _____ m

Show your work.

- 3.
- Card Flips.**
- Nadia wrote
- $-7/8$
- on a card. She flipped it to its opposite, then flipped that result to its opposite. What number does she have now?

Hint Do one flip at a time and mark where each lands.**Answer** _____

- 4.
- Club Balance.**
- A club balance dropped to
- -18.6
- dollars in March. In April it changed to the opposite amount. What is the April balance?

Hint Same distance from 0, other side.

Show your work.

Answer _____ dollars

- 5.
- Missing Start.**
- A number's opposite is
- -4.8
- . What number did you start with?

Hint Flip back: take the opposite of -4.8 .

Show your work.

Start _____

- 6. Ladder and Pit.** A ladder top reaches 6.75 feet above a shop floor (0). A pit bottom sits at the opposite value.
- a.** Write the number for the pit bottom. _____ ft
- b.** How far is each one from the floor? _____ ft

Show your work.

- 7. Twice Over.** What number is the opposite of the opposite of -6.35 ?

Hint Two flips land where you started.

Show your work.

Answer _____

- 8. Three Readings.** Three sensors read -4.5 , 4.5 and -0.5 degrees.

- a.** Write the opposite of each reading. _____
- b.** Which original reading is closest to 0? _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

- 9. Priya's Work.** Priya wrote: "The opposite of the opposite of 8.6 is -8.6 , because the second opposite makes the number negative." Explain her mistake and give the correct answer.

TEST PRACTICE

choose the best answer

- 10.** $-(-3.8)$ equals –
- A** -3.8 **B** 3.8 **C** 0 **D** -0.38
- 11.** The opposite of $-5/6$ is –
- A** $-5/6$ **B** $6/5$ **C** $5/6$ **D** $-6/5$

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

-6.35 C	7/8 Q	-4.8 X	-0.5 T	18.6 F	4.8 L	7.25 R	-7/8 E	-6.75 E	-18.6 Z
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#2b	#3	#4	#5	#6a	#7	#8b
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CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____